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Empowered by AI: exploring the link between literacy, self-efficacy, and career expectations

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ABSTRACT

Artificial intelligence is reshaping higher education, yet little is known about how students' AI related beliefs connect to their career expectations in emerging economy systems. This study examines links between AI literacy, perceived usefulness of AI, attitudes toward AI, AI self-efficacy, and career expectations in AI-driven industries among Vietnamese undergraduates. A cross-sectional survey of 850 students across five campuses was analysed using Partial Least Squares Structural Equation Modelling within an integrated Technology Acceptance Model and Social Cognitive Career Theory framework. AI literacy significantly influenced perceived usefulness, attitudes, and AI self-efficacy, while perceived usefulness was the strongest impact on both attitudes and self-efficacy. Together, literacy, usefulness, and attitudes explained 47.1% of the variance in AI self-efficacy, which in turn was the strongest effect on career expectations in AI-driven industries, accounting for 37.4% of their variance. These findings show that students' confidence in using AI is a key psychological mechanism linking AI literacy and perceived usefulness to career expectations for working in AI-intensive environments. The study provides large-scale evidence from a non-Western context and suggests that higher education institutions should embed structured AI literacy pathways, authentic AI-supported learning tasks, and explicit attention to responsible AI practices in order to strengthen both academic confidence and students' career expectations in AI-driven industries.

1. Introduction

Artificial intelligence (AI) is rapidly transforming higher education by introducing adaptive learning platforms, intelligent tutoring systems, and automated assessment tools that reshape teaching, learning, and administrative operations (Lachheb et al., 2025; Nguyen et al., 2025). While bibliometric analyses document an exponential growth in AI-related scholarship since 2010, most studies remain technology-focused, and student perspectives are underrepresented. Large-scale surveys in Europe and North America report widespread undergraduate use of AI, boosting efficiency and creativity, but also highlight enduring concerns around academic integrity and reduced student–faculty interaction (Stöhr et al., 2024; Strzelecki, 2024). In the United Arab Emirates, nearly 80% of students leverage AI for independent study and collaborative work, yet institutional policies guiding such use are often unclear, underscoring the need for coherent governance (Swidan et al., 2025).

Despite these advances, three critical gaps persist. First, much existing research is siloed by discipline or region whether graphic design majors in Jordan (Alsswey et al., 2025) or U.S. nursing cohorts (Evert Jr., 2024), limiting the transferability of findings (Oyelere & Aruleba, 2025). Second, cross-sectional surveys dominate the field, whereas longitudinal and mixed-methods designs are needed to capture how students' AI perceptions and practices evolve during their academic journeys (Bhatia et al., 2024; Marrone et al., 2024). For example, qualitative work with Chinese undergraduates suggests that generative AI may enhance motivation even as it raises concerns about overreliance and diminished critical thinking (Wang et al., 2024). Third, few studies link students' core AI competencies and attitudes to the motivational construct of AI self-efficacy, or further to concrete career outcomes in AI-driven industries despite growing calls from employers and policymakers for graduates who combine technical proficiency with ethical awareness (Barus et al., 2025; Grimes et al., 2023).

In Vietnam, the Education 4.0 initiative explicitly elevates digital and AI literacy as cornerstones of higher-education reform (Anh et al., 2019; Le, 2020; Le et al., 2025; Le & Ho, 2025). Leading Vietnamese

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universities have begun to embed AI throughout their curricula, research, and institutional processes (Quy et al., 2023), while national studies highlight AI's transformative potential in areas such as recruitment and logistics (Khoa, 2024; McDonald, 2024). Yet empirical data remain scarce on Vietnamese undergraduates' AI literacy, their judgments of AI's usefulness, the formation of attitudes, and the development of AI self-efficacy, and how these factors jointly shape career expectations in AI-intensive fields.

To address these gaps, the present study poses four research questions:

(1) What are Vietnamese undergraduates' perceived usefulness of AI in higher education contexts?
(2) What attitudes do Vietnamese undergraduates hold toward the application of AI?
(3) How are Vietnamese undergraduates' AI literacy, perceived usefulness of AI and attitude toward AI associated with their AI self-efficacy?
(4) How does AI self-efficacy among Vietnamese undergraduates relate to their career expectations in AI-driven industries?

Theoretically, this study contributes by integrating the Technology Acceptance Model and Social Cognitive Career Theory to explain how AI literacy, perceived usefulness of AI, attitudes toward AI, and AI self-efficacy jointly shape AI-related career expectations in higher education (Davis, 1989; Lent et al., 1994). By positioning AI literacy as a foundational antecedent that informs both technology acceptance beliefs and AI-specific self-efficacy, rather than treating it as a background characteristic or outcome, the model extends prior work on students' engagement with AI. Methodologically, this study contributes a large multisite survey of 850 undergraduates across five campuses and employs Partial Least Squares Structural Equation Modelling to test a multi-construct model that spans technology acceptance outcomes and career development orientations, supported by a rigorously validated instrument that can be adapted for comparative research in other systems. Contextually, this study contributes one of the first large-scale, model-based examinations of AI literacy, AI self-efficacy, and AI-related career expectations among Vietnamese undergraduates within the Education 4.0 reform agenda, addressing the current scarcity of empirical evidence from non-Western, emerging economy settings and providing actionable insights for institutional strategy and national policy on AI in higher education.

2. Literature review

2.1. Theoretical framework

To investigate how students engage with AI and how this engagement shapes their academic and career outcomes, this study synthesizes the Technology Acceptance Model (TAM) (Davis, 1989) with Social Cognitive Career Theory (SCCT) (Lent et al., 1994). This combined framework links students' AI related competencies and beliefs about the usefulness of AI to their self-beliefs and expectations about AI intensive careers within a single explanatory model.

2.1.1. Technology Acceptance Model (TAM)

TAM provides a well validated framework for explaining technology uptake in educational contexts. Its central construct, perceived usefulness (PU), reflects users' judgments of how effectively a technology enhances their performance (Venkatesh & Bala, 2008). In this study, PU anchors our examination of undergraduates' evaluations of AI tools in teaching, learning, and administrative processes. Beyond PU, this study introduces a comprehensive AI literacy construct to represent students' technical knowledge, hands-on experience, and critical understanding of AI technologies (Ng et al., 2023 and Syed et al., 2025). In many previous applications TAM has been used primarily to determine acceptance of specific educational technologies without linking these acceptance processes to academic or career outcomes. By embedding AI literacy within the TAM structure, the present study conceptualises literacy as an antecedent that shapes both PU and attitudes toward AI, rather than treating it as a distal background characteristic or as a downstream outcome of technology use.

2.1.2. Social Cognitive Career Theory (SCCT)

SCCT extends TAM by integrating self-efficacy beliefs and outcome expectations as drivers of academic and vocational behavior (Lent et al., 1994). AI self-efficacy, students' confidence in their ability to learn, apply, and critically assess AI tools, serves as a pivotal mediator between their perceptions of AI and their career aspirations in AI-driven industries. Empirical studies have underscored that higher self-efficacy and positive outcome expectations affect greater engagement with AI technologies and a stronger orientation toward AI-related career paths (Swidan et al., 2025; Wang et al., 2024). However, much of this work has focused on general academic self-efficacy or broad STEM career intentions. The present study foregrounds AI specific self-efficacy and AI related career expectations and situates these constructs within an integrated model that connects technology acceptance processes to career development outcomes.

By linking TAM's PU with AI literacy and SCCT's motivational constructs, our model delineates a full pathway from foundational knowledge and perceived value, through attitudes and self-efficacy, to students' career expectations, understood as beliefs about the importance of AI skills, and career readiness, understood as self-assessed preparedness for AI-intensive roles. This linkage clarifies how AI literacy and PU operate as psychological mechanisms that support both day-to-day academic engagement with AI tools and longer-term orientation toward AI-intensive work environments. In Vietnam's dynamic higher education environment, shaped by Education 4.0 reforms that emphasise digital competencies, this dual framework offers both practical and theoretical insights. It illuminates how Vietnamese undergraduates cultivate AI competencies, form judgments about AI's utility, develop confidence in their AI skills, and envision their place in AI-driven labor markets. Furthermore, it extends the applicability of TAM and SCCT to emerging economy settings, demonstrating that AI literacy and career self-efficacy operate in concert to shape technology acceptance and career readiness. At the same time, the integrated TAM and SCCT model has clear boundaries that inform the interpretation of the findings. It focuses on rational, belief driven processes and on AI related self-efficacy, and does not incorporate structural influences such as institutional power, curriculum design or labour market inequalities that also shape AI intensive career trajectories. The framework also privileges positive psychological mechanisms, including perceived usefulness, favorable attitudes and self-efficacy, while conceptually important negative reactions such as AI related anxiety, technostress and dependency are not measured in the present study, even though recent research indicates that these states can accompany intensive AI use among students (Curnell, 2025; Tierney et al., 2025; Wang et al., 2024). For reasons of parsimony, the structural model specifies predominantly linear relations among AI literacy, perceived usefulness, attitudes, AI self-efficacy, and career expectations, which simplifies the more complex, possibly threshold-based dynamics suggested by emerging qualitative work. These limitations delineate the claims that can be made on the basis of the model and indicate the need for future studies that incorporate structural constraints, negative affective responses, and potential nonlinear associations when examining the links between AI literacy and career development.

2.2. AI adoption and research trends in higher education

Studies on the incorporation of Artificial Intelligence (AI) in higher education have markedly increased, especially following the extensive implementation of generative models such as ChatGPT in late 2022

(Lachheb et al., 2025; Stöhr et al., 2024). A thorough bibliometric evaluation indicates a significant increase in publications related to AI applications, including adaptive learning systems, intelligent tutoring, automated assessment, and conversational agents (Lachheb et al., 2025). Across this body of work, there is broad convergence that AI-supported tools can enhance personalized learning, feedback cycles, and student engagement in both online and campus-based settings (Nguyen et al., 2025; Wang et al., 2024). At the same time, most studies focus on short-term learning outcomes or usage intentions, and only a small fraction explicitly examines how AI engagement relates to students' confidence, self-efficacy, or longer-term career orientations. Despite this growth, empirical studies specifically addressing students' experiences and perceptions remain scarce, accounting for only approximately 6% of total research output (Lachheb et al., 2025). This indicates a persistent imbalance between technical or design-oriented scholarship and work that foregrounds student perspectives.

Research on the pedagogical value of AI consistently reports gains in efficiency, scaffolding and individualised learning pathways. For example, studies in China and other contexts show that generative AI can support motivation and cognitive engagement through tailored prompts and explanations, yet they also report tension between these benefits and fears of dependence on automated support and reduced independent thinking (Alsswey et al., 2025; Wang et al., 2024). In parallel, mixed method and conceptual studies emphasise that AI supported teaching and learning can promote creativity, flexibility and access, but raise questions about the depth of higher order thinking and critical reflection when students rely heavily on AI generated outputs (Bhatia et al., 2024; Kasneci et al., 2023). Together, these findings point to a pattern of convergence on short term gains in engagement and convenience, alongside divergence regarding AI's implications for deeper cognitive development.

The implementation of AI technologies in higher education presents significant issues with ethical considerations and governance. Key concerns encompass academic integrity, data privacy, equity, openness, and algorithmic bias (Barus et al., 2025; Tierney et al., 2025). Concerns over academic integrity and plagiarism have escalated, as generative AI tools create greater chances for misbehaviour, necessitating urgent demands for explicit norms, stringent ethical standards, and effective misuse detection systems (Barus et al., 2025; Eke, 2023; Ibrahim, 2023; Lund et al., 2025). Across systems, there is a growing consensus that institutional policies and assessment design must adapt to the presence of generative AI, although specific regulatory responses differ and remain contested. Some institutions adopt restrictive approaches that emphasise detection and prohibition, while others experiment with guided integration and co-creation, which generates variation in how students experience governance and trust surrounding AI (Bannister et al., 2023; Pikhart & Al-Obaydi, 2025; Slimi et al., 2025).

Cross-sectional studies further highlight varying attitudes toward AI across demographic and disciplinary contexts. A large-scale survey of Swedish students found significant differences in chatbot usage and perceptions based on gender and academic discipline, with students from technological fields exhibiting more favorable attitudes than their counterparts in the humanities and medicine, who displayed increased skepticism and apprehension (Stöhr et al., 2024). Comparative studies in other regions identify similar divides between early adopters, who emphasise productivity and support, and more cautious groups, who foreground ethical risks and the potential erosion of academic integrity (Slimi et al., 2025; Strzelecki, 2024). These cross-national patterns suggest that enthusiasm for AI's pedagogical potential is widespread, but its acceptance is mediated by disciplinary cultures, professional norms and prior experiences with digital technologies.

Methodologically, the literature remains dominated by cross-sectional survey designs and small-scale case studies, which constrain understanding of how students' AI perceptions, literacy and self-efficacy evolve (Bhatia et al., 2024; Marrone et al., 2024). Longitudinal and mixed method studies are still rare, and there is limited work that links constructs such as AI literacy, perceived usefulness and attitudes to psychological outcomes and career-related expectations, particularly in non-Western and emerging economy systems. This gap is especially visible in research that extends beyond adoption to examine how AI engagement shapes readiness for AI-intensive labour markets.

In regional terms, scholarship from Europe and the Gulf has paid considerable attention to governance frameworks, academic integrity and institutional policy, while work from East and Southeast Asia has more often focused on classroom integration, learner perceptions and digital competence development. Studies in China and neighbouring systems document rapid uptake of AI-supported learning tools and strong policy pressure for digital transformation, but most concentrate on usage patterns and learning outcomes rather than on AI literacy and career trajectories (Chai et al., 2021; Ng et al., 2023; Wang et al., 2024). Vietnamese contributions to this literature remain limited and scattered, often centred on broad digital transformation rather than student-level AI constructs (Le, 2020; Le & Ho, 2025; Quy et al., 2023). The present study, therefore, responds to both a conceptual and regional gap by examining AI literacy, perceived usefulness, attitudes, self-efficacy, and career expectations in a Vietnamese higher education context.

2.3. Students' perceptions and experiences of AI tools

Recent research has documented the widespread awareness and use of AI tools among university students, indicating both enthusiasm and apprehension in many circumstances. Large-scale cross-national surveys indicate that students perceive significant pedagogical potential in AI applications, particularly chatbots such as ChatGPT, due to their efficiency, responsiveness, and capacity to personalize learning (Stöhr et al., 2024; Swidan et al., 2025). For example, over 50% of nearly 6000 Swedish students acknowledged the educational advantages of chatbots, and a large majority of students in the United Arab Emirates reported using AI for independent study and collaborative work, often motivated by perceived efficiency gains and immediate feedback (Stöhr et al., 2024; Swidan et al., 2025). Across these contexts, perceived benefits cluster around improved access to explanations, faster feedback cycles, and greater flexibility in managing academic tasks, which reinforces the relevance of perceived usefulness as a central construct in models of AI engagement.

At the same time, students express significant ethical and integrity related concerns. In the UAE, high adoption rates are tempered by ambiguity surrounding institutional policies, fears of plagiarism and worries about privacy violations (Swidan et al., 2025). Swedish students report uncertainty about the reliability of AI-generated content and its appropriate use in assessment (Stöhr et al., 2024). Qualitative and survey-based studies in other settings similarly highlight anxieties about academic dishonesty, data protection, and the long-term implications of AI for fairness in evaluation (Eke, 2023; Ibrahim, 2023; Lund et al., 2025; Tierney et al., 2025). These findings converge on the need for clear institutional guidelines and transparent governance, yet they also reveal variation in how students interpret rules and responsibilities across national and institutional contexts. Some cohorts emphasise opportunity and autonomy, while others stress risk and dependency, suggesting the importance of contextual factors in shaping perceptions of AI's trustworthiness and legitimacy in higher education.

Disciplinary and cultural variations further influence students' trust and usage patterns. A comparative study across disciplines in Africa revealed notable differences in trust and acceptance, with engineering students displaying greater comfort and reliance compared to their peers in other disciplines (Oyelere & Aruleba, 2025). Graphic design students in Jordan reported enhanced utilitarian and hedonic benefits from AI integration into their curriculum, including improvements in user experience and learnability, yet they simultaneously expressed reservations about originality and authenticity of AI-generated outputs (Alsswey et al., 2025). These patterns resonate with research from Europe and North America that identifies clusters of AI advocates, cautious critics and passive users, differentiated by disciplinary identity, prior digital experience and perceived alignment between AI tools and field specific norms (Kasneci et al., 2023; Strzelecki, 2024). While there is broad agreement that AI can support learning, the degree of trust and willingness to integrate AI into core disciplinary practices differs markedly.

Qualitative research enriches these findings with more nuanced accounts of cognitive and motivational risks. Nursing students in the United States appreciate AI's support in improving information retention and enhancing learning outcomes, but raise critical concerns regarding its potential to weaken clinical reasoning skills due to over-reliance (Curnell, 2025). Qualitative evidence from Chinese undergraduates indicates that AI-assisted learning offers both motivational and cognitive benefits, yet concurrently raises concerns about potential declines in critical thinking skills and the potential for algorithmic biases embedded in AI systems (Wang et al., 2024). Tierney et al. (2025) highlight the complexity of student viewpoints, revealing a conflict between favorable perceptions of AI-enhanced learning experiences and significant concerns regarding data privacy, reduced creativity, and academic integrity. Across these studies, a common theme is that intensive or uncritical dependence on AI may crowd out opportunities for independent analysis, disciplined practice, and reflective judgment, even when students recognise clear short-term advantages.

To sum up, these quantitative and qualitative findings indicate that increased familiarity with and confidence in AI tools can coexist with emerging risks of overreliance and the normalisation of AI-supported shortcuts in learning. Evidence from contexts such as Sweden, the United Arab Emirates, the United States, and China suggests that moderate levels of AI engagement may enhance efficiency, feedback, and motivation, whereas intensive or unregulated use can reduce opportunities for independent problem solving, critical reflection, and disciplined practice (Stöhr et al., 2024; Swidan et al., 2025; Curnell, 2025; Wang et al., 2024; Tierney et al., 2025). Conceptually, these patterns can be interpreted as potential threshold or inverted U relationships, in which AI literacy and AI self-efficacy support learning and career development up to a certain point, after which very high levels, particularly in weakly regulated environments, may undermine motivation or skill development. For the purposes of the present study, this literature supports the decision to model positive, linear pathways from AI literacy to perceived usefulness, attitudes, self-efficacy, and career expectations, while also acknowledging that future work should examine nonlinear dynamics and the role of institutional regulation in balancing benefits and risks. From a pedagogical perspective, these anxieties about privacy, plagiarism, bias, and overreliance indicate that responsible use and ethical awareness should be treated as core learning outcomes within AI literacy programmes and should be woven through teaching, assessment, and career preparation activities.

2.4. Regional perspectives on AI in higher education: East and Southeast Asia

Scholarship on AI in higher education in East and Southeast Asia reflects both shared drivers and distinct national trajectories. In China, policy initiatives that promote AI as a strategic technology have stimulated rapid uptake of AI supported learning environments and a growing body of research on students' perceptions, motivation and personalized learning experiences (Wang et al., 2024). In neighbouring systems such as Korea and Thailand, governments have also framed AI as central to digital transformation, and higher education institutions have begun to explore AI-enhanced teaching, learning analytics, and smart campus initiatives, although empirical studies remain unevenly distributed across disciplines and institutions. In Vietnam, Education 4.0 policies and institutional strategies emphasise digital and AI literacy as key graduate attributes (Le, 2025; Le & Ho, 2025; Quy et al., 2023). However, empirical work on student-level AI constructs, including AI literacy, perceived usefulness, self-efficacy, and career expectations, is still limited, and much of the existing literature focuses on broader digital transformation or specific platforms such as MOOCs or learning management systems.

Table 1 summarizes selected patterns across four national contexts. It highlights differences in policy framing, research focus, and outcome variables, and shows where AI literacy, psychological constructs, and career-related outcomes have been incorporated. The Vietnam row includes the present study, which addresses gaps by linking AI literacy, perceived usefulness, attitudes, self-efficacy, and AI-related career expectations within a single structural model.

2.5. Hypothesis development and proposed research model

AI literacy refers to an individual's capacity to critically evaluate, understand, and proficiently interact with artificial intelligence (AI) technology, incorporating ethical issues, technical capabilities, and cognitive abilities (Ng et al., 2024). Previous literature consistently underscores the positive impact of AI literacy on users' perceptions and attitudes toward AI. For instance, De Leon et al. (2025) and Syed et al. (2025) affirm that higher AI literacy significantly enhances the perceived usefulness of AI tools. Similarly, empirical evidence highlights a direct correlation between AI literacy and favorable attitudes toward AI applications in educational contexts (Al-Abdullatif, 2024; Galindo-Domínguez et al., 2024). Furthermore, studies indicate that students with robust AI literacy levels display higher AI self-efficacy, reflecting greater confidence in their abilities to learn and apply AI (Bećirović et al., 2025; Bewersdorff et al., 2024; Chai et al., 2021). Based on these findings, the following hypotheses are proposed:

H1. : AI literacy will positively associate with the perceived usefulness of AI.

H2. : AI literacy will positively associate with the attitude toward AI.

H3. : AI literacy will positively associate with AI self-efficacy.

2.5.1. Perceived usefulness of AI

The notion of perceived usefulness, defined as the extent to which students believe that utilising AI technologies improves their academic achievement, has been demonstrated to affect students' attitudes and self-efficacy toward AI tools substantially. Anthony et al. (2022) and Ma et al. (2024) discovered robust empirical evidence suggesting that students who regard AI tools as advantageous and pertinent cultivate more favorable views toward their incorporation in educational settings. Furthermore, perceived usefulness has been demonstrated to bolster students' confidence in the effective utilisation of these technologies; for instance, students who acknowledge the advantages of learning management systems (LMS) or AI tools exhibit heightened confidence and self-efficacy in their application (Al-Qaysi et al., 2020; Chai et al., 2021). As a result, the subsequent hypotheses are formulated:

H4. : Perceived usefulness of AI will positively associate with the attitude toward AI.

H5. : Perceived usefulness of AI will positively associate with the AI self-efficacy.

2.5.2. Attitude toward AI

Students' attitudes toward AI significantly influence their confidence in effectively engaging with AI tools, known as AI self-efficacy. Chai et al. (2021) highlight that students with favorable attitudes toward AI demonstrate increased confidence in understanding and applying AI-related knowledge. Supporting these findings, Bewersdorff et al. (2024) emphasise that positive student attitudes significantly influence AI self-efficacy, independent of students' prior experience or technical abilities. Thus, the subsequent hypothesis is posited:

H6. : Attitude toward AI will positively associate with the AI self-

Table 1
Regional studies on AI in higher education in East and Southeast Asia.

Country	Key policy or system focus	Typical samples and level	Main constructs or models	Primary outcome focus	Identified gaps
China	National strategies that promote AI for innovation and education reform	Undergraduate students across various disciplines, often at research intensive universities	Student perceptions of AI supported learning, AI literacy, motivation, sometimes framed within TAM or related models	Usage patterns, perceived usefulness, learning engagement and motivation, concerns about bias and dependence	Limited focus on career expectations and long term professional trajectories linked to AI literacy and self-efficacy
Korea	Government emphasis on AI competencies, smart campuses and digital transformation in higher education	University students and pre service teachers in technology rich programmes	Technology acceptance, digital competence, attitudes toward AI in teaching and learning	Behavioural intentions to use AI tools, perceived preparedness to integrate AI in professional practice	Few studies explicitly integrate AI literacy, self-efficacy and career outcomes in a single model
Thailand	Policy attention to digital transformation and flexible, technology enhanced learning in higher education	Undergraduate and postgraduate students in blended and online programmes	Digital skills, readiness for AI supported learning, sometimes linked to blended learning or e learning frameworks	Student satisfaction, engagement, readiness for AI enhanced learning environments	Limited incorporation of AI specific literacy constructs and psychological outcomes such as AI self-efficacy and career expectations
Vietnam	Education 4.0 agenda and institutional strategies that foreground digital and AI literacy	Undergraduate students in multisite private and public universities	Digital transformation, AI adoption, AI literacy, and related constructs; the current study integrates TAM and SCCT	AI literacy, perceived usefulness, attitudes, AI self-efficacy, and career expectations in AI-driven industries	Prior work has rarely examined AI literacy and AI self-efficacy as joint effects on AI-related career expectations; the present study addresses this gap using a large multisite sample and a structural model grounded in TAM and SCCT

efficacy.

2.5.3. AI self-efficacy

AI self-efficacy, which denotes individuals' confidence in their capacity to learn and apply AI, is a crucial determinant of career aspirations in AI-driven sectors. Li et al. (2025) demonstrate a strong positive correlation between students' AI self-efficacy and their career aspirations in AI-related domains. Students possessing elevated AI self-efficacy are inclined to establish ambitious professional objectives, demonstrating a heightened readiness to engage in AI-related professions. Therefore, the hypothesis is established as follows:

H7. : AI self-efficacy will positively associate with career expectations in AI-driven industries.

2.6. Conceptual boundaries and possible nonlinear effects

The hypotheses in this study specify linear and positive associations among AI literacy, perceived usefulness of AI, attitudes toward AI, AI self-efficacy and career expectations in AI-driven industries. This specification is consistent with dominant empirical patterns in the AI in education literature and reflects the use of cross-sectional survey data and a parsimonious structural model (Al-Abdullatif, 2024; Chai et al., 2021; De Leon et al., 2025; Syed et al., 2025). At the same time, prior research on students' experiences with AI tools points to processes that may be nonlinear in practice. Studies that document overreliance on AI-supported feedback, weakened clinical reasoning and concerns about diminished critical thinking suggest that very high levels of AI literacy and AI self-efficacy can increase the risk of dependency and may reduce intrinsic motivation for effortful learning (Curnell, 2025; Tierney et al., 2025; Wang et al., 2024). The present study does not estimate curvilinear or threshold effects, and the findings should therefore be interpreted as capturing average linear trends across the sample. Future longitudinal or experimental work could extend this model by testing nonlinear relations explicitly and by examining how institutional policies and learning design moderate the balance between beneficial and potentially detrimental consequences of intensive AI engagement.

2.6.1. Proposed research model

The suggested research model (refer to Fig. 1) incorporates one

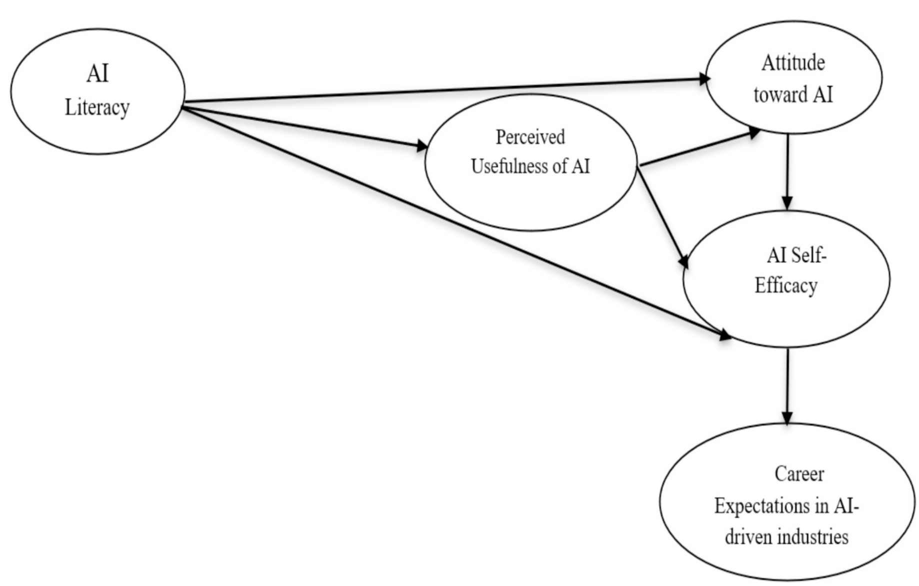

Fig. 1. Proposed research model.

exogenous variable, AI literacy (LI), and four endogenous variables: perceived usefulness of AI (PU), attitudes toward AI (ATT), AI self-efficacy (SE), and career expectations in AI-driven industries (CE). Within this model, AI literacy (LI) influences perceived usefulness (PU), attitudes toward AI (ATT), and AI self-efficacy (SE). Additionally, perceived usefulness (PU) affects both attitudes toward AI (ATT) and AI self-efficacy (SE). Attitudes toward AI (ATT) subsequently influence AI self-efficacy (SE), and finally, AI self-efficacy (SE) impacts career expectations in AI-driven industries (CE).

3. Methodology

3.1. Research instrument

A structured survey instrument was created to systematically collect comprehensive data in accordance with the stated theoretical framework and hypotheses. The instrument was explicitly created to implement and quantitatively evaluate essential constructs delineated in the proposed study paradigm (refer to Fig. 1). The poll was conducted among undergraduate students from five campuses of the participating higher education institution, guaranteeing diverse representation. The survey tool consists of six separate sections, each assessing a particular construct pertinent to the research aims. Sections 2, 3, 4, and 6 employ a uniform five-point Likert scale from 1 (Strongly Disagree) to 5 (Strongly Agree), enhancing comparability and simplifying analysis across categories including perceived utility, views toward AI, and career aspirations. Section 5, which evaluates AI self-efficacy, utilizes a five-point confidence scale from 1 (Not Confident) to 5 (Extremely Confident), precisely capturing respondents' confidence levels concerning AI proficiency.

The survey instrument was theory-driven, adapted from established and validated scales rather than developed independently. Perceived usefulness and attitudes toward AI were drawn from the Technology Acceptance Model (TAM), with items contextualized for AI-supported teaching, learning, and assessment. Measures for AI self-efficacy and career expectations were grounded in Social Cognitive Career Theory (SCCT), modified to reflect AI-specific skills and career readiness. Additionally, AI literacy items were informed by contemporary empirical frameworks to capture students' conceptual understanding and practical experience. To ensure cross-cultural and content validity, all items underwent a rigorous process of translation, back-translation, expert review, and pilot testing. While reworded for the higher education context, the items maintain the theoretical logic of their original instruments. The scales were empirically re-validated using PLS-SEM, demonstrating robust internal consistency, convergent validity, and discriminant validity. For transparency and replication, the complete instrument is provided in Appendix 1.

3.2. Sampling procedures and data collection

Given the complexity and diversity of the student population across multiple campuses, this study employed a systematic multistage cluster sampling strategy, ensuring both methodological rigor and representativeness. The sampling process was structured into three clearly defined stages. In the first stage, all five campuses of the participating higher education institution, located in Hanoi, Ho Chi Minh City, Can Tho, Da Nang, and Binh Dinh, were selected as primary clusters due to their comprehensive academic programs that explicitly integrate AI-related coursework or practical modules. These campuses represent diverse geographic and educational contexts, enhancing the study's generalizability. The second stage involved a purposeful selection of academic programs within each campus. These programs were systematically identified based on their explicit inclusion of AI components, operationally defined as offering at least one mandatory course, module, or project focusing on AI concepts, technologies, or applications. This criterion was established to precisely reflect the diverse levels of undergraduate students' exposure to AI, thereby ensuring the relevance and applicability of the gathered data. In the third stage, a stratified random sampling technique was employed to select individual students from the enrollment lists within the designated AI-integrated programs. Stratification was based on major, academic year, and gender to ensure proportional representation and reduce potential sampling biases. Continuous demographic monitoring throughout the sampling process ensured the sample's representativeness and addressed potential disparities or imbalances.

In addition, several procedural remedies were implemented to minimise the risk of common method bias, in accordance with the recommendations of Rodríguez-Ardura and Meseguer-Artola (2020). Survey data were collected across multiple campuses, with data collection staggered over different time periods for each campus. Furthermore, the questionnaire was intentionally kept concise and anonymous, and all items were phrased clearly and unambiguously to reduce respondents' cognitive burden and response bias.

As with many large-scale survey studies, this research is subject to potential selection bias arising from voluntary participation. Because respondents self-selected into the study, students with greater interest in AI, higher digital confidence, or more positive attitudes toward technology may have been more likely to participate. Conversely, students who are less confident, less engaged with AI, or more skeptical may be underrepresented. This self-selection mechanism can lead to systematic differences between participants and non-participants, thereby limiting the representativeness of the sample.

The survey underwent a pre-testing phase with 60 participants to validate the robustness of the instrument. Internal consistency was confirmed, as Cronbach's alpha remained within the acceptable range (>0.70) as stated by Hair et al. (2010). Based on the pilot test, this instrument was utilized for the formal data collection.

Data collection occurred in May and June of the 2025 academic year. Ethical issues were meticulously upheld, with participation being voluntary and dependent on obtaining informed consent, following express approval from the institution's research ethics committee (Decision No. 758/QĐ-ĐHFPT, dated July 4, 2025). Before accessing the survey, all participants were presented with an informed consent statement detailing the study's purpose, the voluntary nature of participation, the scope of the questions, and the estimated completion time. Participants were explicitly informed of their right to decline or withdraw at any point. Consent was obtained electronically; only those who provided active agreement could proceed to the questionnaire. To ensure data privacy, no personally identifiable information, such as names or student IDs, was collected. Responses were recorded via Google Forms, and data were reported only in aggregate. Access to the raw dataset was restricted to the research team, who had the password to view. These data were used exclusively for research purposes and were not shared with any external parties.

The poll was conducted electronically using Google Forms. To enhance participation rates and accessibility, survey invitations were distributed through various institutional communication channels, including official university emails, instructor assistance, and social media sites such as Facebook and Zalo. The structured multistage sampling methodology effectively balances logistical feasibility with methodological precision, facilitating the collection of comprehensive and representative data that authentically reflects undergraduate students' experiences and perspectives regarding AI across diverse educational settings.

3.3. Participants

The study distributed a formal survey to 1200 undergraduates across five campuses: Hanoi, Can Tho, Ho Chi Minh City, Binh Dinh, and Danang. Data collection took place between May 1 and June 30, 2025, resulting in 850 valid responses, a 71% response rate. A summary of the participant demographics is provided in Table 2.

Table 2
Demographic information.

Variable	Category	Frequency (N = 850)	%
Gender	Male	359	42.2
	Female	479	56.4
	Prefer not to say	12	1.4
Year of study	1st year	84	9.9
	2nd year	223	26.2
	3rd year	246	28.9
	4th year	297	34.9
Degree Program	Business Administration	418	49.2
	Digital Art and Design	82	9.6
	Information Technology	264	31.1
	Language	44	5.2
	Multimedia Communication	42	4.9
Campus	Binh Dinh	17	2.0
	Can Tho	177	20.8
	Da Nang	31	3.6
	Hanoi	291	34.2
	Ho Chi Minh	334	39.3

Within this sample, 56.4% identified as female, 42.2% as male, and 1.4% preferred not to disclose their gender, providing a balanced gender distribution. Students spanned all undergraduate levels, with 9.9% in their first year, 26.2% in the second year, 28.9% in the third year, and 34.9% in the fourth year.

Respondents came from five majors: Business Administration (49.2%), Information Technology (31.1%), Digital Art and Design (9.6%), Language Studies (5.2%), and Multimedia Communication (4.9%). This disciplinary mix reflects both STEM and non-STEM fields, allowing for analysis of AI perceptions across varied academic contexts. Geographically, the largest student cohorts were based at the Ho Chi Minh City (39.3%) and Hanoi (34.2%) campuses, with smaller contributions from Can Tho (20.8%), Da Nang (3.6%), and Binh Dinh (2.0%).

While the single-institution nature of our sample limits full generalizability to all Vietnamese universities, the multisite design captures diverse urban and regional settings, program types, and student levels. These features support cautious extrapolation of findings to similar multisite or regional university systems within Vietnam.

3.4. Data analysis

Data analysis utilized Partial Least Squares Structural Equation Modelling (PLS-SEM) through SmartPLS 3.0 software. In accordance with the systematic process proposed by Hair et al. (2018), the analysis consisted of two consecutive phases: (1) evaluation of the measurement model and (2) assessment of the structural model. In the preliminary phase, the measurement model was assessed to confirm the reliability and validity of the survey instrument, which consisted of 27 validated questions measuring the five constructs illustrated in the proposed study model (refer to Fig. 1). Prior methodological comparisons showed that CB-SEM often required dropping indicators to achieve an acceptable model fit, potentially weakening content validity. In contrast, PLS-SEM enables the retention of theoretically meaningful indicators while achieving strong composite reliability, convergent validity (AVE), and discriminant validity (HTMT), aligning with the study's emphasis on construct coverage and interpretability (Hair et al., 2017). In the meantime, the data of this study were cross-sectional, self-reported, and collected across multiple campuses and disciplines, which might introduce deviations from multivariate normality, although the sample size was large (N = 850). Hair et al. (2017) stated that PLS-SEM did not impose strict distributional assumptions and was therefore more robust for this type of educational survey data than CB-SEM.

Reliability assessment in this study emphasised internal consistency. While Cronbach's alpha has conventionally been used to evaluate scale reliability based on inter-item correlations (Hair et al., 2010), this approach has limitations in the context of PLS-SEM. Accordingly, composite reliability (CR) was employed to assess internal consistency, as it provides a more appropriate and robust estimate for variance-based structural equation modelling (Hair et al., 2018).

The validity examination was divided into convergent and discriminant validity. Convergent validity was assessed by analysing factor loadings and Average Variance Extracted (AVE) values, confirming that all constructs achieved sufficient convergence, with AVE values exceeding the established threshold of 0.50 (Hair et al., 2018). Discriminant validity was evaluated using a thorough methodology, which included cross-loadings, the Fornell–Larcker criterion, and the Heterotrait-Monotrait (HTMT) ratio criterion, thereby affirming the distinctiveness of the constructs.

All key reliability and validity indicators are displayed in Table 3 below.

In the next phase, the structural model was evaluated to examine the proposed links among constructs. This entailed examining path coefficients to ascertain the direction and magnitude of relationships, coefficients of determination (R^2) to assess the explanatory power of endogenous constructs, predictive relevance (Q^2) to verify the model's predictive efficacy, and effect size (f^2) to measure the practical significance of each structural path. This meticulous analytical method guaranteed thorough testing of the submitted hypotheses and yielded extensive insights into the structural relationships within the research model.

4. Results

4.1. Common method bias

Common method bias refers to variance attributable to the measurement method rather than the construct supposedly represented by the measures (Podsakoff et al., 2003). In this study, the researchers used a statistical control technique, Harman's single-factor test (Harman, 1976), to determine whether common method bias was an issue in the present study. As recommended by Podsakoff et al. (2003), an exploratory factor analysis (EFA) with extraction fixed at a single factor was performed. This analysis included all 18 items across the five latent constructs: AI literacy, AI self-efficacy, perceived usefulness of AI, attitude toward AI, and career expectation in AI-driven industries. The results indicated that the single factor explained 38.179% of the variance (as shown in Table 4), which is below the recommended threshold of 50% (Podsakoff et al., 2003). Consequently, common method bias does not pose a significant concern for this dataset.

4.2. Testing the measurement model

To evaluate the reliability and validity of the measurement model, outer loadings, Composite Reliability (CR), and Average Variance Extracted (AVE) were examined, with the results presented in Table 5. Most items (>0.70) indicate their suitability as measures of the respective variables (Hair et al., 2018). Only two items (PU7 and SE1)

Table 3
Summary of guidelines for measurement model analysis.

Assessment	Guidlines
Internal consistency reliability	Composite reliability (CR) CR should be higher than 0.70 (CR > 0.70)
Convergent validity	The average variance extracted (AVE) AVE should be higher than 0.50 (AVE > 0.50)
Discriminant validity	Fornell-Larcker criterion The square root of the AVE of each latent construct should be greater than its highest correlational value among the constructs Heterotrait-Monotrait (HTMT) criterion The value should not be exceeded: HTMT.85 value of 0.85 HTMT.90 value of 0.90

Table 4
Results of Harman's single factor analysis.

Component	Initial Eigenvalues			Extraction sums of squared loadings		
	Total	% of variance	Cumulative %	Total	% of variance	Cumulative %
1	6.872	38.179	38.179	6.872	38.179	38.179
2	1.471	8.172	46.350			
3	1.178	6.544	52.894			
4	1.085	6.029	58.924			
5	0.788	4.377	63.301			
6	0.734	4.078	67.380			
7	0.679	3.773	71.153			
8	0.652	3.620	74.773			
9	0.609	3.384	78.156			
10	0.584	3.242	81.398			
11	0.574	3.189	84.588			
12	0.512	2.842	87.430			
13	0.449	2.497	89.927			
14	0.440	2.446	92.373			
15	0.412	2.287	94.660			
16	0.367	2.039	96.699			
17	0.360	1.999	98.698			
18	0.234	1.302	100.000			

Extraction Method: Principal Component Analysis.

Table 5
Internal consistency reliability and convergent validity.

Construct	Items	Mean (M)	Standard deviation (SD)	Factor loading	Composite reliability (CR)	Average variance extracted (AVE)
Perceived Usefulness of AI (PU)	PU1:AI can assist teachers by providing real-time answers to students' questions.	3.65	0.857	0.701	0.905	0.579
	PU2: AI can help teachers manage their time more efficiently.	3.67	0.828	0.744		
	PU3: AI can provide personalized support to different groups of students.	3.65	0.841	0.811		
	PU4: AI can personalize lessons based on individual student needs.	3.71	0.823	0.832		
	PU5: AI can provide universal access to education for students with special needs.	3.73	0.805	0.793		
	PU7: AI can reduce human errors in grading.	3.51	0.871	0.647		
	PU8: AI can provide constant, personalized feedback to each student.	3.66	0.850	0.781		
Career Expectations in AI-driven Industries (CE)	CE1: AI-related skills will be important for my future career.	3.84	0.877	0.860	0.855	0.663
	CE2: My university is preparing me well to work in environments where AI is used.	3.60	0.850	0.747		
	CE4: Learning how to collaborate with AI tools will be essential for my success.	3.91	0.845	0.832		
Attitude toward AI (ATT)	ATT2: AI will help solve major problems in education.	3.54	0.985	0.756	0.839	0.635
	ATT5: AI increases the efficiency of time management and risk reduction tasks.	3.7	0.866	0.809		
	ATT7: AI will contribute positively to global economic growth.	3.75	0.891	0.824		
AI Literacy (LI)	LI1: I feel well informed about the concept of artificial intelligence (AI).	3.06	0.788	0.833	0.777	0.636
	LI4: I have experience using a wide range of AI tools over the past year.	3.02	0.797	0.760		
AI Self-Efficacy (SE)	SE1: Selecting and using AI tools (e.g., ChatGPT, automated grading).	3.5	0.777	0.620	0.770	0.531
	SE3: Learning new AI-related skills when needed.	3.68	0.817	0.813		
	SE4: Identifying the strengths and limitations of AI applications.	3.45	0.901	0.740		

have outer loadings below 0.70, which may marginally diminish convergent validity; however, they remain acceptable given robust CR and AVE values (Hulland, 1999). The measuring model exhibits acceptable internal consistency and convergent validity for all constructs. The Composite Reliability (CR) ratings for all constructs exceed the well-recognized criterion of 0.70, indicating robust internal consistency (Hair et al., 2018). The Perceived Usefulness of AI exhibits the highest construct reliability at 0.905, followed by Career Expectations in AI-driven Industries at 0.855, Attitude toward AI at 0.839, AI Literacy at 0.777, and AI Self-Efficacy at 0.770. Regarding convergent validity, all constructs exhibit Average Variance Extracted (AVE) values beyond the minimum acceptable criterion of 0.50 (Hair et al., 2018), with Career Expectations in AI-driven Industries reflecting the highest convergent validity (AVE = 0.663). The AVE scores for the other constructions vary from 0.531 to 0.636, affirming the sufficiency of the individual items in reflecting their respective constructs. The combination of satisfactory outer loadings, robust composite reliability (CR), and average variance extracted (AVE) indicates the internal consistency reliability and convergent validity of the measurement model.

Table 6 reports the corrected item–total correlations for all measurement items across the five constructs. According to Nunnally and Bernstein (1994), a corrected item–total correlation of 0.30 or higher indicates that an item contributes adequately to its underlying construct and demonstrates acceptable internal consistency. Based on the 0.30 criterion proposed by Nunnally and Bernstein (1994), the majority of items demonstrate acceptable corrected item–total correlations, supporting the internal consistency of the scales. Items falling slightly below the threshold of 0.30, including LI1 and LI4, were subsequently evaluated using more stringent criteria in the PLS-SEM measurement model, where decisions regarding item retention or removal were made based on factor loadings and composite reliability above.

Discriminant validity was assessed using three methods: the Fornell-Larcker Criterion, the Cross Loadings Criterion, and the Heterotrait-Monotrait (HTMT) Criterion, as detailed in Tables 7, 8, and 9, respectively.

Table 7 indicates that the measuring model meets the standards for discriminant validity according to the Fornell–Larcker method. Hair et al. (2018) assert that the square root of the AVE for each construct exceeds its correlations with all other constructs. The square root of AVE for AI Literacy is 0.797, surpassing its connections with AI Self-Efficacy (0.406), Attitude toward AI (0.253), Career Expectations in AI-driven Industries (0.294), and Perceived Usefulness of AI (0.278). Comparable trends are seen across all other constructs in Table 7. Career Expectations in AI-driven Industries exhibit the highest discriminant value of 0.814. The inter-construct correlations, which range from 0.253 to 0.632, indicate that the constructs are empirically distinct. In summary, these findings validate that the constructs included in the investigation exhibit sufficient discriminant validity, satisfying the Fornell–Larcker Criterion and bolstering the reliability of the measurement model.

To enhance the establishment of discriminant validity, the cross-loadings of all measuring items were analysed in Table 8. Per the cross-loading criterion established by Hair et al. (2018), each item must demonstrate a greater loading on its designated construct than on any other construct. Table 8 indicates that all products fulfil this criterion. Items assessing Attitude toward AI (ATT2, ATT5, and ATT7) exhibit substantial loadings on their designated construct, ranging from 0.756 to 0.824, all surpassing their cross-loadings with alternative latent variables. Likewise, the components associated with Career Expectations in AI-driven Industries (CE1, CE2, and CE4) exhibit significant loadings on their respective construct (e.g., CE1 = 0.860) relative to other constructs. The AI Literacy indicators, LI1 and LI4, show the most significant loadings on their respective constructs, with values of 0.833 and 0.760, respectively. The measures assessing Perceived Usefulness of AI (PU1 to PU8) consistently exhibit higher loadings on their designated construct than on any alternative, with notably strong values such as PU4 = 0.832 and PU3 = 0.811. The AI Self-Efficacy construct is effectively exemplified by SE1, SE3, and SE4, each demonstrating much higher loadings on their respective construct compared to others (e.g., SE3 = 0.813 vs. 0.616). The cross-loading analysis results affirm that discriminant validity is firmly established, hence reinforcing the structural integrity of the measurement model.

Table 6
Items- total statistics.

Construct	Items	Corrected item-total correlation
Perceived Usefulness of AI (PU)	PU1	0.602
	PU2	0.628
	PU3	0.719
	PU4	0.746
	PU5	0.687
	PU7	0.541
	PU8	0.692
Career Expectations in AI-driven Industries (CE)	1CE	0.645
	2CE	0.491
	4CE	0.584
Attitude toward AI (ATT)	ATT2	0.511
	ATT5	0.519
	ATT7	0.562
AI Literacy (LI)	LI1	0.274
	LI4	0.274
AI Self-Efficacy(SE)	SE1	0.32
	SE3	0.433
	SE4	0.358

The examination of discriminant validity through the Heterotrait-Monotrait (HTMT) ratio indicated that almost all values resided within an acceptable range, specifically between 0.449 and 0.89, as illustrated in Table 9. These values remain significantly below the widely endorsed criterion of 0.90 (Henseler et al., 2015), signifying that the conceptions are empirically distinct. The HTMT value of 0.715 between Attitude toward AI and Perceived Usefulness of AI, along with the value of 0.459 between AI Literacy and Attitude toward AI, offers compelling evidence of discriminant validity. These results affirm that each construct encapsulates a distinct facet of the theoretical framework and exhibits minimal correlation with other constructs in the model. Nevertheless, an exception is noted in the HTMT value between AI Self-Efficacy and Career Expectations in AI-driven Industries, which is marginally above the threshold at 0.913. Although this may immediately provoke apprehension, Henseler et al. (2015) propose that HTMT levels nearing, yet not surpassing, 1.0 do not inherently signify a deficiency in discriminant validity, especially under specific circumstances. Particularly, when (1) the measurement indicators exhibit substantial and consistent loadings, and (2) the sample size is comparatively big, heightened HTMT values may nevertheless be deemed appropriate. The outside loadings in the present investigation vary from 0.70 to 0.86, signifying robust item reliability. The sample size of 850 individuals is adequate to mitigate the possibility of illusory HTMT inflation. Collectively, these criteria indicate that the HTMT value of 0.913, albeit somewhat exceeding the standard, does not undermine the overall discriminant validity of the measurement model.

4.3. *Testing the structural model*

Fig. 2 depicts the structural model outcomes derived from the Partial Least Squares Structural Equation Modelling (PLS-SEM) investigation. The model has five latent constructs: AI Literacy, Perceived Usefulness of AI, Attitude toward AI, AI Self-Efficacy, and Career Expectations in AI-driven Industries, each assessed by numerous reflecting indicators.

Table 10 reports R^2 and Q^2 for each endogenous construct. AI Self-Efficacy has the strongest explanatory power ($R^2 = 0.471$), meaning that 47.1% of its variance is explained by the model. Career Expectations ($R^2 = 0.374$) and Attitude toward AI ($R^2 = 0.336$) are also well accounted for. In contrast, perceived usefulness of AI exhibited a low coefficient of determination ($R^2 = 0.077$), indicating that AI literacy explains only a small proportion of variance in students' usefulness judgments.

Hair et al. (2018) state that the Q^2 values for all constructs are positive, signifying that the model possesses acceptable predictive accuracy. AI Self-Efficacy ($Q^2 = 0.246$), Career Expectations ($Q^2 = 0.245$), and Attitude toward AI ($Q^2 = 0.209$) exhibit moderate predictive significance. Nonetheless, the Perceived Usefulness of AI exhibits a comparatively low Q^2 of 0.045 in relation to the Q^2 of the other constructs.

Table 11 illustrates the effect sizes (f^2) of the predictor constructs on the endogenous variables, elucidating the relative contribution of each predictor within the structural model. Cohen's (1988) criteria indicate that f^2 values of 0.02, 0.15, and 0.35 correspond to minor, medium, and high (substantial) effect sizes, respectively. The impact size analysis indicates that AI Self-Efficacy exerts the most significant influence on Career Expectations in AI-driven Industries ($f^2 = 0.598$), signifying a considerable effect. Likewise, Perceived Usefulness of AI significantly influences both Attitude toward AI ($f^2 = 0.409$) and AI Self-Efficacy (f^2

Table 7
Results of discriminant validity of Fornell-Larcker criterion.

Construct	AVE	AI literacy	AI self-efficacy	Attitude toward AI	Career expectations in AI-driven industries	Perceived usefulness of AI
AI literacy	0.636	0.797				
AI self-efficacy	0.531	0.406	0.729			
Attitude toward AI	0.635	0.253	0.481	0.797		
Career expectations in AI-driven industries	0.663	0.294	0.612	0.502	0.814	
Perceived usefulness of AI	0.579	0.278	0.632	0.571	0.569	0.761

Table 8
Results of discriminant validity of cross-loadings criterion.

Items	AI literacy (LI)	AI self-efficacy (SE)	Attitude toward AI (ATT)	Career expectations in AI-driven industries (CE)	Perceived usefulness of AI (PU)
ATT2	0.185	0.351	0.756	0.346	0.383
ATT5	0.232	0.412	0.809	0.379	0.479
ATT7	0.185	0.383	0.824	0.469	0.494
CE1	0.268	0.516	0.442	0.86	0.481
CE2	0.182	0.455	0.31	0.747	0.392
CE4	0.263	0.521	0.464	0.832	0.51
LI1	0.833	0.368	0.187	0.272	0.239
LI4	0.76	0.275	0.221	0.192	0.204
PU1	0.214	0.378	0.392	0.416	0.701
PU2	0.242	0.498	0.494	0.471	0.744
PU3	0.229	0.492	0.44	0.439	0.811
PU4	0.23	0.497	0.451	0.452	0.832
PU5	0.226	0.557	0.457	0.467	0.793
PU7	0.149	0.418	0.356	0.33	0.647
PU8	0.181	0.501	0.432	0.437	0.781
SE1	0.38	0.62	0.288	0.285	0.337
SE3	0.317	0.813	0.4	0.616	0.429
SE4	0.222	0.74	0.354	0.388	0.604

= 0.289). Conversely, AI Literacy exhibits minimal to moderate effects on Perceived Usefulness ($f^2 = 0.084$) and AI Self-Efficacy ($f^2 = 0.095$), along with a negligible effect on Attitude toward AI ($f^2 = 0.014$). Furthermore, Attitude toward AI exerts a minor effect on AI self-efficacy ($f^2 = 0.027$). The findings indicate that Perceived Usefulness and AI Self-Efficacy are the most significant constructs in the model. Although Cohen's conventional thresholds provide a useful reference point, the substantive meaning of effect sizes in this study depends on their theoretical role within the integrated TAM–SCCT framework. Smaller effects associated with AI literacy reflect its function as a foundational enabling condition, whereas larger effects linked to AI self-efficacy highlight its proximal and motivational role in shaping career expectations.

Table 12 summarizes the findings from hypothesis testing for the structural model illustrated in Fig. 2. All seven presented hypotheses received robust support, evidenced by statistically significant path coefficients ($p < 0.01$) and elevated t-values as per Hair et al. (2018). AI Literacy impacts three primary constructs: it positively affects Perceived Usefulness of AI ($\beta = 0.278$, $p = 0.001$), Attitude toward AI ($\beta = 0.102$, $p = 0.001$), and AI Self-Efficacy ($\beta = 0.235$, $p = 0.001$), thereby affirming its essential role in influencing students' perceptions and confidence regarding AI. However, the associated effect size is negligible ($f^2 = 0.014$), falling below Cohen's (1988) threshold for a minor effect. This indicates that while AI literacy has a detectable association with students' attitudes toward AI, its practical contribution is minimal when considered relative to other constructs in the model.

The most significant correlation was identified between AI Self-Efficacy and Career Expectations in AI-driven Industries ($\beta = 0.612$), indicating that students' confidence in their AI talents is a major effect on their career aspirations. Furthermore, the Perceived Usefulness of AI markedly improves both Attitude toward AI ($\beta = 0.543$) and AI Self-Efficacy ($\beta = 0.484$), indicating that students who regard AI as beneficial are likely to exhibit a more favorable attitude and increased confidence in its practical use. Finally, Attitude toward AI exerts a positive, albeit lesser, influence on AI Self-Efficacy ($\beta = 0.145$) in comparison to the effect of Perceived Usefulness of AI on AI Self-Efficacy ($\beta = 0.484$). These findings together corroborate the model's framework and underscore the interrelated roles of students' literacy, perception, and attitude in influencing their trust in AI and career expectations associated with AI.

The mediation analysis demonstrates that all indirect effects within the model are statistically significant, with acceptable *p*-values and bias-corrected confidence intervals that exclude zero, thereby confirming robust mediation pathways. AI self-efficacy emerges as the primary and most consistent mediator, linking AI literacy and the perceived usefulness of AI to career expectations in AI-driven industries. In contrast, attitude toward AI serves a secondary, less pronounced mediating role. Of all evaluated pathways, the indirect effect of perceived usefulness on career expectations—mediated by AI self-efficacy—is the most substantial with $\beta = 0.344$. This suggests that students' recognition of AI's practical value is the most potent driver of their confidence and, by extension, their AI-related career aspirations. While AI literacy contributes indirectly by shaping both attitudes and self-efficacy, its impact is comparatively modest. This reinforces the perspective that literacy acts as a foundational enabling condition rather than a primary motivational driver (Table 13).

5. Discussion

5.1. Theoretical implications

This study examined Vietnamese undergraduates' AI literacy, perceived usefulness of AI, attitudes toward AI, AI self-efficacy, and career expectations in AI-driven industries. By integrating the Technology Acceptance Model and Social Cognitive Career Theory, we tested four research questions that considered how AI literacy and perceived usefulness of AI influence attitude toward AI, how attitude toward AI and perceived usefulness of AI influence AI self-efficacy, how AI literacy,

Table 9
Results of discriminant validity of Heterotrait-Monotrait (HTMT) criterion.

	AI literacy	AI self-efficacy	Attitude toward AI	Career expectations in AI-driven industries	Perceived usefulness of AI
AI literacy					
AI self-efficacy	0.849				
Attitude toward AI	0.459	0.751			
Career expectations in AI-driven industries	0.51	0.913	0.68		
Perceived usefulness of AI	0.449	0.89	0.715	0.699	

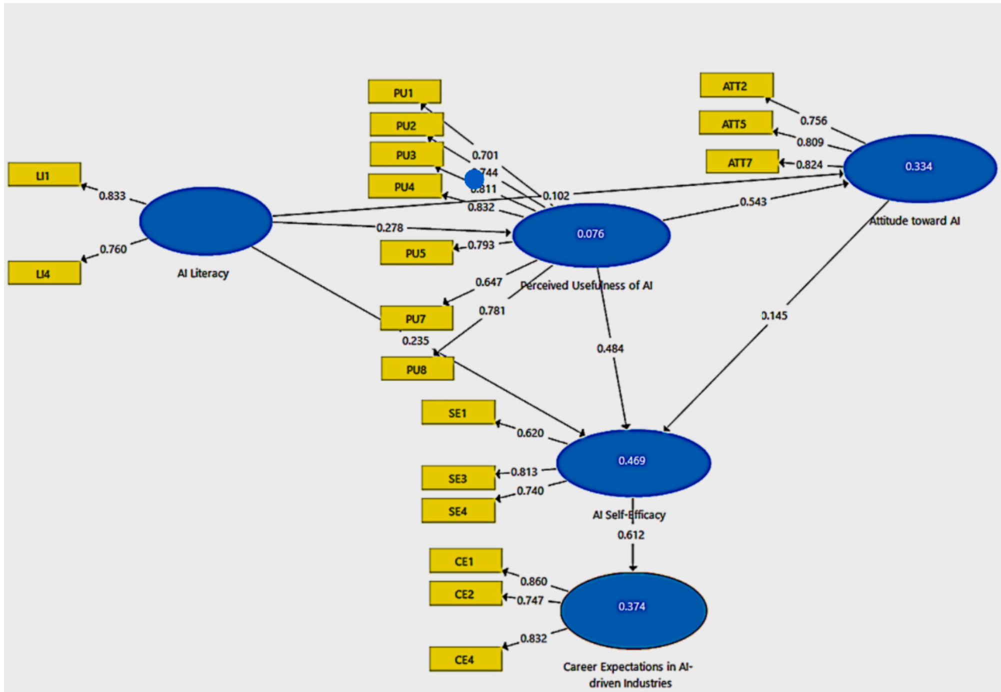

Fig. 2. Results of PLS-SEM analysis.

Table 10
Coefficient of determination (R^2) and predictive relevance (R^2).

Construct	R^2	Q^2
AI self-efficacy	0.471 (47.1%)	0.246
Attitude toward AI	0.336 (33.6%)	0.209
Career expectations in AI-driven industries	0.374 (37.4%)	0.245
Perceived usefulness of AI	0.077 (7.7%)	0.045

perceived usefulness of AI, and attitude toward AI jointly contribute to AI self-efficacy, and how AI self-efficacy relates to career expectations and readiness. Taken together, the findings refine the integrated TAM and SCCT framework. Perceived usefulness of AI, rather than AI literacy or attitude toward AI, has the most significant impact on AI self-efficacy, and AI self-efficacy was the primary driver of students' AI-related career expectations in this context.

The study finds that both AI literacy and perceived usefulness of AI are independently associated with attitude toward AI. However, the perceived usefulness of AI exerts roughly three times the impact of AI literacy ($f^2 = 0.409$ vs. $f^2 = 0.014$). Although AI literacy provides foundational knowledge and exposure, perceived usefulness represents students' *direct evaluation of whether AI improves their learning and performance*. This appraisal is more immediate and motivationally salient than general awareness or familiarity. Students may understand AI in principle, but confidence and positive attitudes emerge primarily when they experience AI as concretely beneficial in completing academic tasks. In line with Anthony et al. (2022) and Ma et al. (2024), this underscores that clear, task-relevant benefits are the primary lever for shifting student mindsets, which refines TAM by highlighting the central motivational role of perceived usefulness in the context of generative AI. By contrast, the measure of AI literacy, comprised of conceptual knowledge items and self-reported tool usage, explained only a small share of attitudinal variance. This suggests that familiarity with AI terminology and tools, in the absence of meaningful academic applications, contributes weakly to attitude formation. These findings suggest that AI literacy theory should move beyond treating literacy as a standalone competency. Instead, literacy should be understood as context-dependent and value-activated, with its effects realised through meaningful application. AI literacy gains motivational and developmental significance only when learners can translate knowledge into experiences of usefulness.

Perceived usefulness of AI again emerged as the dominant predictor of AI self-efficacy ($f^2 = 0.289$), echoing Hanham et al. (2021), who highlighted that tangible advantages are crucial for building confidence. AI literacy contributed modestly ($f^2 = 0.095$), indicating that deeper, hands-on skill development rather than abstract familiarity is more critical for fostering belief in one's competence. Attitudes toward AI, while statistically significant, produced a negligible effect ($f^2 = 0.027$). In theoretical terms, this pattern indicates that a positive mindset alone is insufficient. Students need to see and experience AI's value in authentic academic tasks to translate literacy and attitudes into strong efficacy beliefs. The results, therefore, extend TAM by demonstrating that usefulness is a key psychological mechanism linking AI literacy to AI self-efficacy and extend SCCT by positioning perceived usefulness as

Table 11
Effect size (f^2).

Construct	AI literacy	AI self-efficacy	Attitude toward AI	Career expectations in AI-driven industries	Perceived usefulness of AI
AI Literacy		0.095 (Minor to medium effect size)	0.014 (Negligible effect size)		0.084 (Small to medium effect size)
AI Self-Efficacy				0.598 (Substantial effect size)	
Attitude toward AI		0.027 (Minor effect size)			
Perceived Usefulness of AI		0.289 (Medium to substantial effect size)	0.409 (Substantial effect size)		

Table 12
Hypothesis testing results.

Hypothesis paths	Coefficient (β)	T Statistic	*P* values	Results
H1: AI Literacy - > Perceived Usefulness of AI	0.278	7.817	0.001	Suppoted
H2: AI Literacy - > Attitude toward AI	0.102	3.379	0.001	Supported
H3: AI Literacy - > AI Self-Efficacy	0.235	7.59	0.001	Supported
H4: Perceived Usefulness of AI - > Attitude toward AI	0.543	13.36	0.001	Supported
H5: Perceived Usefulness of AI - > AI Self-Efficacy	0.484	13.135	0.001	Supported
H6: Attitude toward AI - > AI Self-Efficacy	0.145	4.266	0.001	Supported
H7: AI Self-Efficacy - > Career Expectations in AI-driven Industries	0.612	22.717	0.001	Supported

part of the cognitive appraisal processes that support efficacy development.

Consistent with SCCT and corroborating Bewersdorff et al. (2024), AI self-efficacy was the strongest driver of career expectations in AI-driven Industries (β = 0.612). This construct alone explained 37.4% of the variance in students' anticipated readiness for AI-driven roles. The findings confirm that domain-specific self-efficacy operates as a central mechanism in career psychology. In this study, it connects technology-related beliefs to expectations about future work in AI-intensive environments in a non-Western, emerging economy system.

The pattern of weaker relationships is also informative. The relatively small effects of AI literacy on attitude toward AI and AI self-efficacy, and the limited role of attitude toward AI in affecting AI self-efficacy, suggest that the links between knowledge, affect, and confidence are more conditional than initially hypothesised. One plausible explanation is that the literacy scale captures breadth of familiarity rather than depth of practice, which may be insufficient to shift attitudes and efficacy in a context where AI is still evolving in curricula. Another possibility is that students' confidence depends more on direct experience of AI's usefulness than on general positivity. These results indicate that the hypothesised paths are not uniformly strong and that future theory testing should distinguish between different dimensions of literacy, such as critical and ethical literacy, and examine how they interact with perceived usefulness of AI and attitude toward AI over time.

The low R^2 for perceived usefulness of AI suggests that students' perceptions of AI's usefulness are not primarily driven by general familiarity or knowledge, but rather by contextual experiences of AI use across specific courses, tasks, and assessment practices. The present finding indicates that AI literacy functions as an enabling condition rather than a sufficient determinant of perceived usefulness of AI.

5.2. *Practical implications*

When AI literacy, perceived usefulness of AI, and attitude toward AI were entered together, they explained 47.1% of the variance in AI self-efficacy, a substantial portion. This suggests that successful institutional interventions need to combine robust, hands-on training that builds technical proficiency, clear benefits demonstrated in core subjects, and supportive messaging that normalises thoughtful experimentation with AI tools. In practical terms, these findings indicate that institutions should sequence AI learning so that students first acquire core literacy, then repeatedly experience AI as genuinely useful in their disciplinary courses, and finally reflect on how these experiences shape their confidence and career plans. The results refine practical interpretations of TAM by showing that signals about "ease of use" embedded in literacy cannot substitute for direct engagement with AI's concrete advantages.

The strong pathway from AI self-efficacy to career expectations in AI-driven Industries has important implications for teacher professional development and curriculum design. Because AI self-efficacy accounts for a large share of the variance in career outlook, Vietnam's Education 4.0 reforms should prioritise experiential learning opportunities that allow students to use AI in realistic projects and to see its relevance for future work. A first concrete intervention is a compulsory first-year module, such as "AI for Study and Work," for all majors, in which students assess their baseline AI literacy, complete guided activities comparing AI-supported and non-AI-supported solutions, and develop a personal plan for using AI in their degree programme. These opportunities include internships with AI-supported workflows, capstone projects that integrate AI tools into problem-solving, and collaborative labs where students can prototype AI-enhanced solutions. A second set of interventions involves redesigning existing disciplinary courses so that at least one assignment per course requires students to specify prompts, generate AI outputs, critique their accuracy and bias, and then combine AI suggestions with their own reasoning in a graded product. A third intervention is to integrate AI-rich career preparation tasks into final-year courses, for example, asking students to analyse job advertisements for AI-related skills in their field, build portfolios showcasing AI-enhanced projects, and rehearse how to present these competencies in interviews. Within AI literacy courses, emphasis should shift from passive definition memorisation toward guided, ethical, and technical tool use, for example, structuring data pipelines, interpreting AI outputs, and addressing bias.

The findings also suggest that teacher professional development should focus on building educators' AI literacy and AI self-efficacy so they can design tasks that make AI's usefulness visible and model responsible use. Workshops that support lecturers in integrating AI into assessment design, feedback processes, and discipline-specific activities are likely to have a multiplier effect on students' perceptions of usefulness and confidence. For instance, staff development programmes can train lecturers to design assessments that award marks for the quality of AI prompts, the critical evaluation of AI outputs, and the explicit justification for when students choose not to use AI. In addition, career services and industry partnerships should explicitly address AI-related competencies, helping students translate classroom experiences with AI into narratives about employability and readiness for AI-intensive roles. Universities can also offer short micro-credential courses or badges in "Responsible AI for Professional Practice" that bundle these

Table 13
Results of mediating analysis.

Indirect effect	Coefficient (β)	Standard deviation (STDEV)	T statistics	P values	Bias corrected confidence intervals	Results
AI literacy - > AI self-efficacy	0.171	0.024	7.009	0.001	0.124, 0.220	Supported
AI literacy - > Attitude toward AI	0.151	0.023	6.529	0.001	0.108, 0.198	Supported
AI literacy - > Career expectations in AI-driven industries	0.249	0.025	9.950	0.001	0.197, 0.296	Supported
Attitude toward AI - > Career expectations in AI-driven industries	0.089	0.022	4.109	0.001	0.050, 0.131	Supported
Perceived usefulness of AI - > AI self-efficacy	0.079	0.018	4.316	0.001	0.440, 0.116	Supported
Perceived USEFULNESS of AI - > Career expectations in AI-driven industries	0.344	0.031	10.937	0.001	0.285, 0.411	Supported

curricular and extracurricular activities and provide students with recognisable evidence of their AI literacy, self-efficacy, and ethical awareness for future employers.

5.3. Ethical implications and responsible AI education

The findings have direct implications for how responsible AI use is conceptualised within AI literacy and AI self-efficacy. In this study, AI literacy is defined to include functional knowledge of tools as well as an understanding of ethical issues, technical capabilities, and limitations. When literacy encompasses awareness of bias, transparency, and potential misuse, higher literacy can support more informed and responsible decision-making rather than simply more intensive use of AI tools. Students who understand how AI systems can misclassify, reproduce stereotypes, or generate inaccurate content are better positioned to question outputs, cross-check sources, and refuse uses of AI that conflict with academic or professional norms.

AI self-efficacy exhibits a similar dual character. Higher self-efficacy may be protective when it reflects confidence in monitoring, adapting, and correcting AI systems, since such students are more likely to challenge unfair or unsafe uses of AI in academic and workplace settings. At the same time, overconfidence that is not grounded in ethical reasoning may increase the likelihood of irresponsible shortcuts, such as uncritical reliance on AI-generated text or the delegation of complex judgment to opaque systems. The strong link between self-efficacy and career expectations observed in this study, therefore, suggests that readiness for AI-intensive careers should be understood to include professional responsibility. Graduates who expect to work with AI need technical skills and a clear sense of accountability, fairness, and the importance of human oversight in AI-supported work.

The literature reviewed in this study highlights persistent concerns about academic integrity, data privacy, and algorithmic bias in higher education (Barus et al., 2025; Eke, 2023; Ibrahim, 2023; Lund et al., 2025; Tierney et al., 2025). Interpreted through a fairness and inclusion lens, these concerns point to the risk that unequal access to high-quality AI tools and uneven ethical guidance could widen existing educational and labour-market gaps. Students in well-resourced programmes with explicit instruction on responsible AI use are more likely to develop ethically informed literacy and self-efficacy. Peers without such support may either avoid AI entirely or use it in ways that compromise integrity and future opportunities. To mitigate these risks, AI literacy curricula should integrate ethical and technical competencies from the outset. Practical measures include embedding case studies that foreground dilemmas such as plagiarism and bias, designing activities in which students detect and correct problematic outputs, and making institutional policies on AI use visible and discussable within courses.

In light of these patterns, the strong positive pathways from AI literacy and perceived usefulness to AI self-efficacy and from self-efficacy to career expectations should not be interpreted as unambiguously beneficial. The exact mechanisms that build confidence and readiness can also, at very high levels and in weakly regulated environments, contribute to overreliance on AI tools and reduced motivation for effortful learning. Institutions, therefore, need to couple efforts to enhance AI literacy and self-efficacy with explicit training in responsible AI practices and with attention to equitable access, so that the benefits of AI-enabled learning and career preparation are distributed fairly and do not exacerbate existing inequalities.

6. Limitations and future research directions

The findings should be interpreted within several design-related and contextual limitations. The study employed a cross-sectional survey, which restricts the ability to draw causal conclusions about the relations among AI literacy, perceived usefulness of AI, attitude toward AI, AI self-efficacy, and career expectations in AI-driven industries. The design also cannot capture how these constructs develop across different stages of students' academic trajectories or in response to changes in institutional policy and labour market conditions. The sampling frame was confined to a single private, multisite university in Vietnam. Although the five campuses are geographically dispersed and include a mix of programmes and student profiles, the results may not generalise to public universities, vocational colleges, or elite research institutions. Patterns of AI access, instructional support, and career orientation may differ in systems with different funding models, governance structures, or student demographics. Although subsequent PLS-SEM analyses supported the adequacy of the AI literacy construct in terms of composite reliability and convergent validity, the limited number of retained indicators and the presence of items with marginal internal consistency restrict the depth with which AI literacy can be captured. In this study, we now explicitly acknowledge this limitation of low predictive relevance (Q^2) values and caution against overinterpreting the model's predictive capability.

Future research should adopt longitudinal and mixed-method designs to examine how AI literacy, perceived usefulness, attitudes, and self-efficacy evolve and how these trajectories relate to shifts in career expectations and actual career outcomes. Panel studies that follow cohorts from entry into university through graduation would allow researchers to model within-person change and to test whether early gains in AI literacy and self-efficacy translate into sustained engagement with AI-supported learning and into concrete employment pathways. Experimental or quasi-experimental designs that evaluate specific AI literacy interventions would also strengthen causal inference.

There is also a need for comparative and cross-cultural studies that extend beyond a single national system. Research that samples across different Vietnamese institutions and East and Southeast Asian systems would enable testing whether the structural relations observed here hold in settings with different policy priorities and resource levels. Such work could draw on the validated instrument used in this study while adapting items to local governance arrangements and labour market demands.

Diverse learner groups remain underrepresented. The present study focused on full-time undergraduates in programmes that already incorporate AI-related content. Future work should include vocational students, part-time learners, adult returners and learners in disciplines with less obvious AI integration. These groups may display different patterns of literacy, perceived usefulness, and self-efficacy, and they may face distinct opportunities and constraints in AI-intensive labour markets.

Cultural and contextual factors are likely to moderate the observed relationships. Norms regarding academic integrity, institutional policies on AI governance, local labour market signals about AI-intensive occupations, and broader societal narratives about automation may all shape how students interpret AI literacy and how they link AI engagement to future careers. Subsequent studies could test moderation by institutional type, discipline, gender, or region and employ multi-group structural equation modelling to examine whether path strengths differ systematically across contexts. Transparency about these limitations enhances the scientific rigor and interpretive caution of the present study and highlights clear priorities for future research.

Developing and validating richer, multi-item AI literacy scales, potentially distinguishing between conceptual, practical, and critical dimensions to deal with two metrics to measure AI literacy. Enhancing these measures would enable a more accurate assessment of how AI literacy influences students' motivation and career paths. Furthermore, it would create a more robust theoretical link between AI literacy, technology acceptance models, and career development theories.

Future research should extend this work by employing longitudinal designs, holdout samples, or PLSpredict procedures to strengthen predictive assessment.

7. Conclusion

This study investigated how AI literacy, perceived usefulness, and attitudes toward AI shape Vietnamese undergraduates' AI self-efficacy and, in turn, their career expectations in AI-driven industries. AI literacy underpinned students' appreciation of AI's practical benefits and contributed to the formation of positive attitudes and confidence in using AI tools, while the perceived usefulness of AI significantly influences AI self-efficacy. AI self-efficacy has the greatest effect on students' expectations for working in AI-intensive environments, confirming the central role of confidence in linking technology-related learning experiences to career psychology in a non-Western higher education system.

AI-driven learning environments, therefore, present a dual landscape. On the one hand, they can enhance feedback, personalisation, and access to complex problem-solving resources, which supports the development of literacy, efficacy, and career readiness. On the other hand, if introduced without adequate guidance and regulation, they may encourage overreliance on automated support, weaken opportunities for independent critical thinking, and amplify existing inequalities in access to high-quality tools and ethical supervision.

Within the methodological and contextual boundaries described in the limitations, the findings suggest that embedding structured AI learning pathways, practice-oriented experiences, internships, capstone projects, and collaborative labs can strengthen both AI literacy and AI self-efficacy and can support more informed and responsible career planning. Future longitudinal and cross-cultural research with more diverse learner groups will be important to test the robustness of these patterns and to examine how responsible AI education can foster equitable and sustainable career development in AI-intensive labour markets. Given the rapid evolution of AI in education, continuous refinement of conceptual frameworks and measurement instruments will be necessary to ensure that future studies capture new forms of AI engagement and their implications for learning and careers.

CRediT authorship contribution statement

Nguyen Thi Thao Ho: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Ha Van Le:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix 1. Survey

Section 1: Demographic Information

Q1. What is your gender?

- Female
- Male
- Non-binary
- Prefer not to say

Q2. What is your year of study?

- Year 1
- Year 2
- Year 3
- Year 4

Q3. What is your major?

- Information Technology
- Business Administration
- Digital Art and Design
- Multimedia Communications
- English Language
- Other (please specify): ________

Q4. Which campus do you attend?

- Hanoi
- Ho Chi Minh City
- Da Nang
- Can Tho
- Binh Dinh

Q5. Have you passed all your exams in the last academic year?

- Yes
- No

Q6. What was your GPA last year? (4.0 scale)

- Below 2.0
- 2.0–2.49
- 2.5–2.99
- 3.0–3.49
- 3.5–4.0

Section 2: AI Literacy

Please indicate your agreement with the following statements:

(Scale: 1 – Strongly Disagree, 2 – Disagree, 3 – Neutral, 4 – Agree, 5 – Strongly Agree)

1. I feel well informed about the concept of artificial intelligence (AI).
2. I regularly use AI tools (e.g., ChatGPT, Grammarly) in my academic work.
3. I have received adequate formal or informal training on AI in education.
4. I have experience using a wide range of AI tools over the past year.

Section 3: Perceived Usefulness of AI

Please indicate your agreement with the following statements:

(Scale: 1 – Strongly Disagree, 2 – Disagree, 3 – Neutral, 4 – Agree, 5 – Strongly Agree)

1. AI can assist teachers by providing real-time answers to students' questions.
2. AI can help teachers manage their time more efficiently.
3. AI can provide personalized support to different groups of students.
4. AI can personalize lessons based on individual student needs.
5. AI can provide universal access to education for students with special needs.
6. AI can automate exam grading.
7. AI can reduce human errors in grading.
8. AI can provide constant, personalized feedback to each student.

Section 4: Attitude toward AI

Please indicate your agreement with the following statements:

(Scale: 1 – Strongly Disagree, 2 – Disagree, 3 – Neutral, 4 – Agree, 5 –

Strongly Agree)

1. AI will lead to some job losses in the future.
2. AI will help solve major problems in education.
3. AI could make services and learning experiences feel less personal.
4. AI technologies are costly and require significant resources.
5. AI increases the efficiency of time management and risk reduction tasks.
6. AI could dominate or strongly influence societal decision-making.
7. AI will contribute positively to global economic growth.

Section 5: AI self-efficacy
Indicate your confidence with the following:
(Scale: 1 – Not confident, 2 – Slightly confident, 3 – Moderately confident, 4 – Confident, and 5 – Extremely confident)

1. Selecting and using AI tools (e.g., ChatGPT, automated grading).
2. Understanding key AI concepts (e.g., machine learning, neural networks).
3. Learning new AI-related skills when needed.
4. Identifying the strengths and limitations of AI applications.

Section 6: Career Expectations in AI-driven industries
Please indicate your agreement with the following statements:
(Scale: 1 – Strongly Disagree to 5 – Strongly Agree)

1. AI-related skills will be important for my future career.
2. My university is preparing me well to work in environments where AI is used.
3. I am concerned that AI might replace jobs in my future industry.
4. Learning how to collaborate with AI tools will be essential for my success.

Data availability

Data will be made available on request.

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